NAME: SAARTHAK SABHARWAL

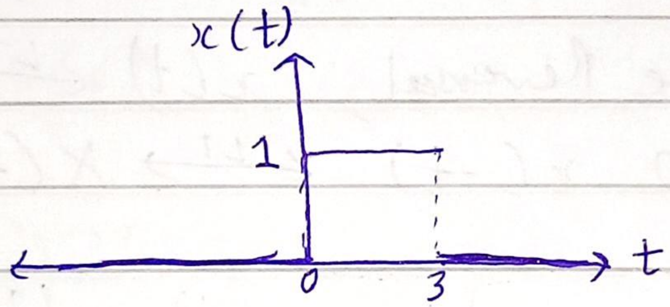
ROLL NO: 2023102055

ECS 101 - Assignment 3

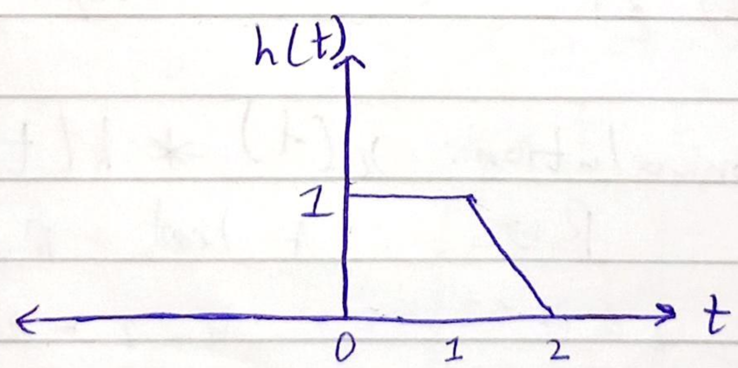
1. a) $x(t) = u(t) - u(t-3)$

$$= \begin{cases} 0, & t<0 \\ 1, & t \geq 0 \end{cases} - \begin{cases} 0, & t-3<0 \Rightarrow t<3 \\ 1, & t-3 \geq 0 \Rightarrow t \geq 3 \end{cases}$$

$$= \begin{cases} 0, & t<0 \\ 1, & 0 \leq t < 3 \\ 0, & t \geq 3 \end{cases}$$



$$h(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 2-t, & 1 < t \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

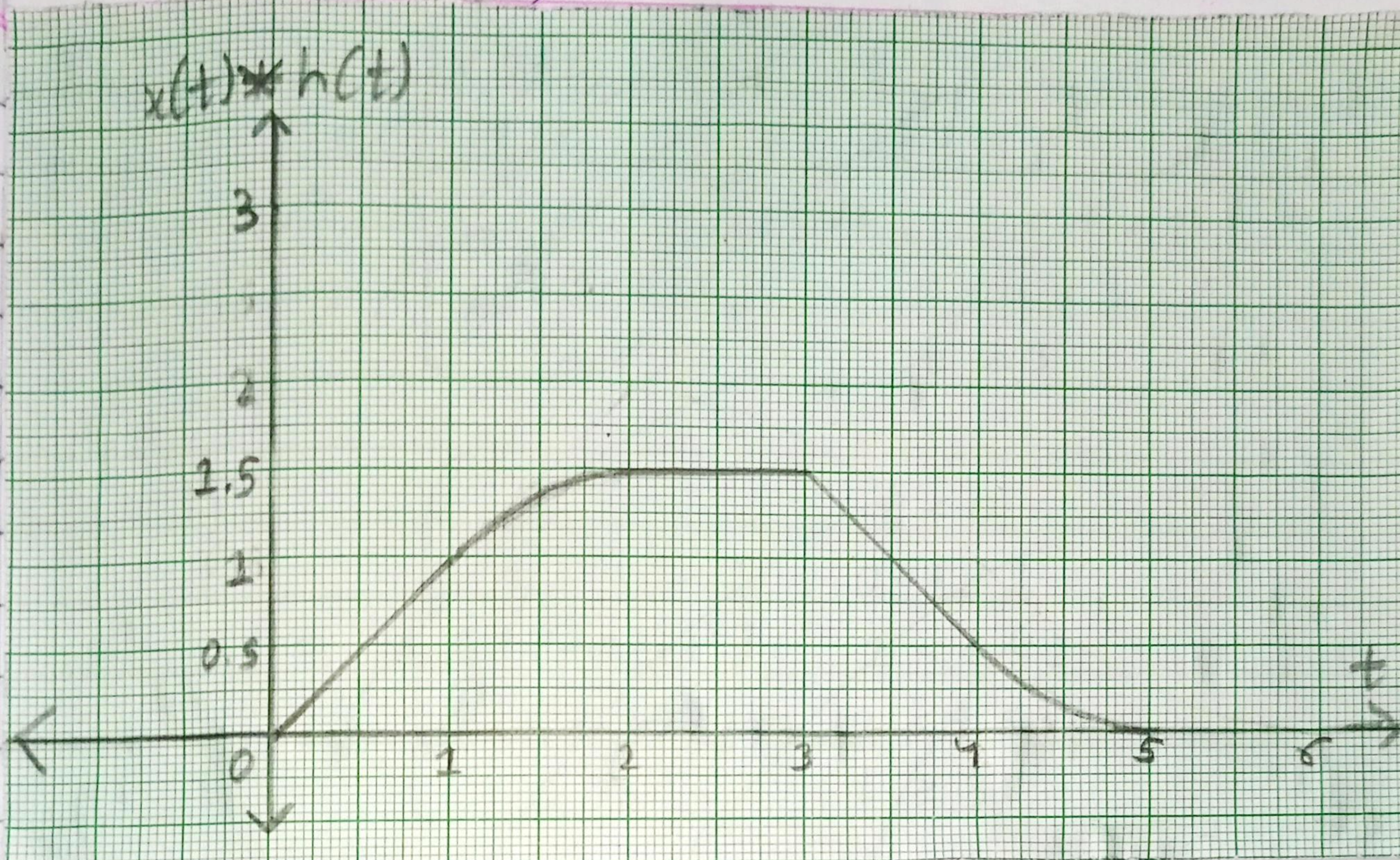


b) $$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{0} \cancel{x(\tau)}^{0}\, h(t-\tau)\, d\tau + \int_{0}^{3} \cancel{x(\tau)}^{1}\, h(t-\tau)\, d\tau + \int_{3}^{\infty} \cancel{x(\tau)}^{0}\, h(t-\tau)\, d\tau$$

$$= \int_{0}^{3} h(t-\tau)\, d\tau = \int_{t-3}^{t} h(\tau)\, d\tau$$

[$\because$ $t - \tau = r \Rightarrow -d\tau = dr$; limits changed & sign-flipped]



$$
= \begin{cases}
\int_{t-3}^{t} 0 \, d\tau, & t < 0 \\
\int_{0}^{t} 1 \, d\tau & 0 \le t \le 1 \\
\int_{0}^{1} 1 \, d\tau + \int_{1}^{t} (2-\tau) \, d\tau, & 1 < t \le 2 \\
\int_{0}^{1} 1 \, d\tau + \int_{1}^{2} (2-\tau) \, d\tau, & 2 < t \le 3 \\
\int_{t-3}^{1} 1 \, d\tau + \int_{1}^{2} (2-\tau) \, d\tau, & 3 < t \le 4 \\
\int_{t-3}^{2} (2-\tau) \, d\tau, & 4 < t \le 5 \\
\int_{t-3}^{t} 0 \, d\tau, & t \ge 5
\end{cases}
$$

$$
= \begin{cases}
0, & t < 0 \\
t, & 0 \le t \le 1 \\
\frac{1}{2}[4t - t^2 - 1], & 1 < t \le 2 \\
3/2, & 2 < t \le 3 \\
\frac{9}{2} - t, & 3 < t \le 4 \\
\frac{1}{2}[t^2 - 10t + 25], & 4 < t \le 5 \\
0, & t \ge 5
\end{cases}
$$

2. $y(t) = x_1(t) * x_2(t) * x_3(t)$

$x_1(at) * x_2(at) * x_3(at) = [x_1(at) * x_2(at)] * x_3(at)$

$= \left[\int_{-\infty}^{\infty} x_1(az)\, x_2(at - az)\, dz \right] * x_3(at)$

Let $az = r \Rightarrow a\,dz = dr$ — ①

$\Rightarrow \left[\frac{1}{a} \int_{-\infty}^{\infty} x_1(r)\, x_2(at - r)\, dr \right] * x_3(at)$, $\boxed{a \neq 0}$

Let $x_1(t) * x_2(t) = c_1(t) = \int_{-\infty}^{\infty} x_1(z) x_2(t-z) dz$

$\Rightarrow \frac{1}{|a|} c_1(at) * x_3(at)$ $[\because \forall a < 0, \text{ limits flip}]$

$= \frac{1}{|a|} \int_{-\infty}^{\infty} c_1(az) \cdot x_3(at - az)\, dz$

~~Let~~ ~~$az = r$~~ Using ①, we have

$\Rightarrow \frac{1}{|a|^2} \int_{-\infty}^{\infty} c_1(r)\, x_3(at - r)\, dr$

Let $c_1(t) * x_3(t) = c_2(t) = \int_{-\infty}^{\infty} c_1(r) x_3(t-r) dr$

$\Rightarrow \frac{1}{a^2} c_2(at)$ (final expression)

Now, $c_2(at) =$ ~~$c_1(at)$~~ $c_1(t) * x_3(t) = x_1(t) * x_2(t) * x_3(t)$

$\Rightarrow c_2(t) = y(t)$ $\Rightarrow c_2(at) = y(at)$

$\therefore x_1(at) * x_2(at) * x_3(at)$

$= \boxed{\frac{1}{a^2} y(at)}$, $a \in \mathbb{R}$, $a \neq 0$.

3. $x(t) = e^{-3|t|} + e^{-|t|}$

a) $x(t) \xleftrightarrow{LT} X(s) = \int_{-\infty}^{\infty} e^{-st} x(t)\, dt$

$\cancel{X_s}$

$\Rightarrow X(s) = \int_{-\infty}^{\infty} \left[e^{-3|t|} + e^{-(|t|)}\right] e^{-st}\, dt$

$$= \int_{-\infty}^{\infty} e^{-3|t|-st}\, dt + \int_{-\infty}^{\infty} e^{-|t|-st}\, dt$$

$$= \int_{-\infty}^{0} e^{3t-st}\, dt + \int_{0}^{\infty} e^{-3t-st}\, dt$$

$$+ \int_{-\infty}^{0} e^{t-st}\, dt + \int_{0}^{\infty} e^{-t-st}\, dt$$

$$= \left[\frac{e^{(3-s)t}}{3-s} + \frac{e^{(1-s)t}}{1-s}\right]_{-\infty}^{0} + \left[\frac{e^{-(3+s)t}}{-(3+s)} + \frac{e^{-(1+s)t}}{-(1+s)}\right]_{0}^{\infty}$$

$$= \frac{1}{3-s} + \frac{1}{1-s} - 0 - 0 + 0 + 0 - \frac{1}{-(3+s)} - \frac{1}{-(1+s)}$$

$$\Rightarrow X(s) = \frac{8(3-s^2)}{(3-s)(3+s)(1-s)(1+s)} = \frac{N(s)}{D(s)}$$

$N(s) = 0 \Rightarrow s^2 = 3 \Rightarrow s = \pm\sqrt{3}$ (zeroes) [O]

$D(s) = 0 \Rightarrow s = \pm 3, \pm 1$ (Poles) [X]

For ROC, $Re(3-s) > 0$ & $Re(1-s) > 0$

& $Re(3+s) > 0$ & $Re(1+s) > 0$

$\Rightarrow Re(s) < 3$ & $Re(s) < 1$ & $Re(s) > -3$ & $Re(s) > -1$

$\Rightarrow$ ROC : $|R(s)| < 1 \Rightarrow -1 < R(s) < 1$

b) i) $$\int_{-\infty}^{\infty} x(t)e^{-t/3}\,dt = X(\cancel{1/3}\; s = 1/3)$$

$$\left[\because \int_{-\infty}^{\infty} e^{-st}x(t)\,dt = X(s) \text{ from (a)}\right]$$

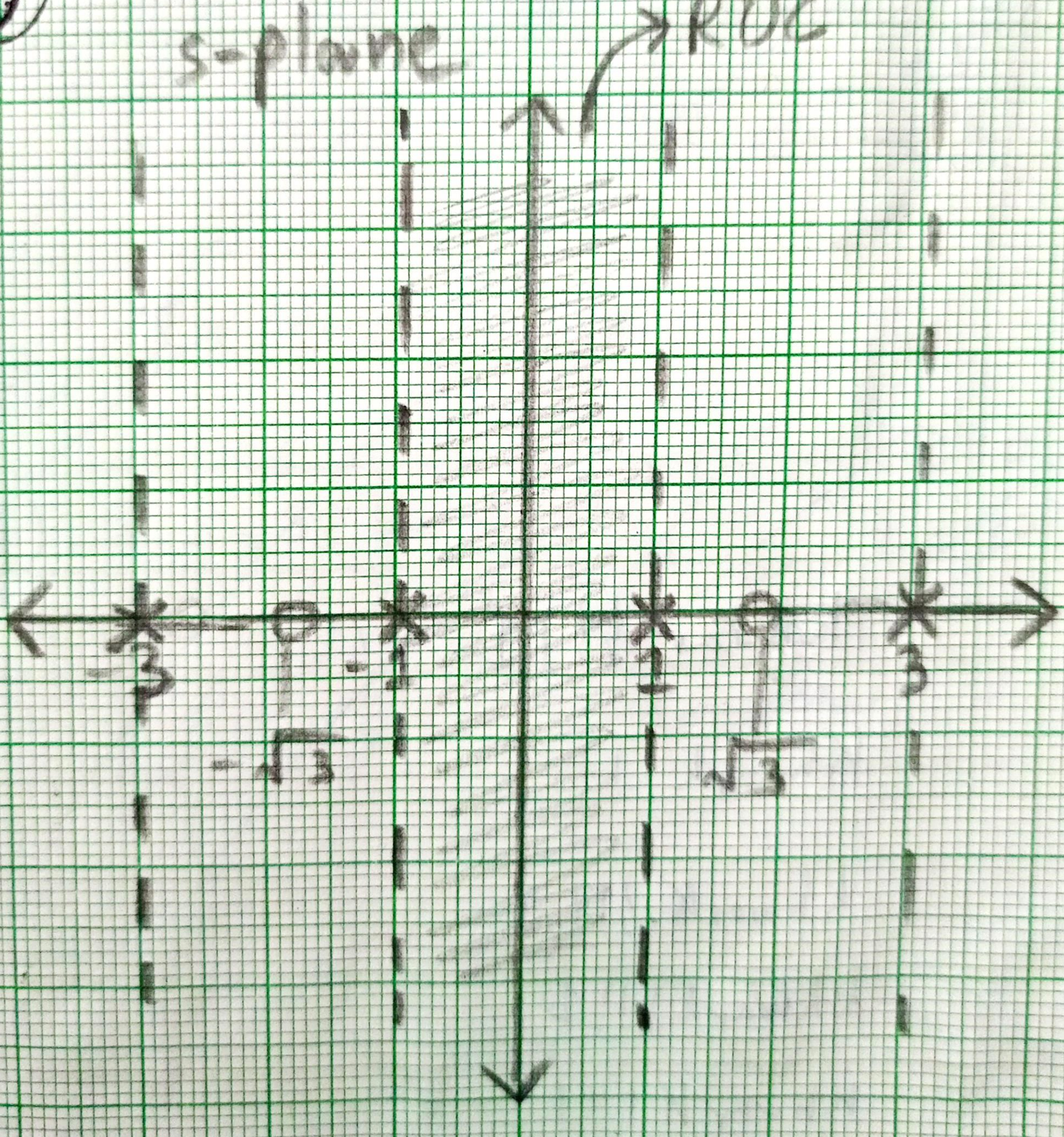
$$\Rightarrow X\left(\frac{1}{3}\right) = \frac{8\left(3-\frac{1}{9}\right)}{\left(9-\frac{1}{9}\right)\left(1-\frac{1}{9}\right)} = \frac{\cancel{8}\times\overset{13}{\cancel{26}}\times 9\times\cancel{9}}{\underset{40}{\cancel{80}}\times\cancel{8}\times\cancel{9}}$$

$$= \frac{117}{40}$$

ii) $$\int_{-\infty}^{\infty} x(t)e^{-j2t}\,dt = X(s = 2j)$$

$$= \frac{8(3-4j^2)}{(9-4j^2)(1-4j^2)} = \frac{8\times 7}{13\times 5} = \frac{56}{65}$$

⑥



4. $x(t) \longrightarrow \boxed{S1} \longrightarrow \begin{cases} x(t), & t \leq 10 \\ 0, & t > 10 \end{cases}$

- $x_1(t) \xrightarrow{S1} y_1(t) = \begin{cases} x_1(t), & t \geq 10 \\ 0, & t < 10 \end{cases}$

  $x_2(t) \xrightarrow{S1} y_2(t) = \begin{cases} x_2(t), & t \geq 10 \\ 0, & t < 10 \end{cases}$

$$\Rightarrow \alpha x_1(t) + \beta x_2(t) \xrightarrow{S1} \begin{cases} \alpha x_1(t) + \beta x_2(t), & t \geq 10 \\ 0, & t < 10 \end{cases}$$

$$= \alpha \begin{cases} x_1(t), & t \geq 10 \\ 0, & t < 10 \end{cases} + \beta \begin{cases} x_2(t), & t \geq 10 \\ 0, & t < 10 \end{cases}$$

$$= \alpha y_1(t) + \beta y_2(t)$$

$\therefore$ S1 is $\boxed{\text{linear}}$

- $x(t) \longrightarrow \boxed{D_\Delta} \longrightarrow x(t-\Delta) \longrightarrow \boxed{S1} \longrightarrow y_1(t)$

$$y_1(t) = \begin{cases} x(t-\Delta), & t \leq 10 \\ 0, & t > 10 \end{cases}$$

$$x(t) \longrightarrow \boxed{S1} \longrightarrow \begin{cases} x(t), & t \leq 10 \\ 0, & t > 10 \end{cases} \longrightarrow \boxed{D_\Delta} \longrightarrow y_2(t)$$

$$y_2(t) = \begin{cases} x(t-\Delta), & t \leq 10+\Delta \\ 0, & t > 10+\Delta \end{cases}$$

$\Rightarrow y_1(t) \neq y_2(t)$

$\therefore$ S1 is $\boxed{\text{time-variant}}$ or not time-invariant.

- S1 is $\boxed{\text{causal}}$, since it requires input only till the current time (and outputs it as is based on t).
  $y(t) = f\{x(\tau), \tau \leq t\}$; $x(t) \longrightarrow \boxed{S1} \longrightarrow y(t)$.

- Let $|x(t)| < B < \infty$, and
  $x(t) \longrightarrow \boxed{S1} \longrightarrow y(t)$.

$$\Rightarrow y(t) = \begin{cases} x(t), & t \leq 10 \\ 0, & t > 10 \end{cases}$$

Now, let us assume that $|y(t)| = \infty$ for some t.

$t > 10$: Not possible, as $y(t) = 0 \quad \forall \ t > 10$.

$t \leq 10$: $y(t) = x(t) \Rightarrow |y(t)| < B < \infty$

which is a contradiction, proving the assumption wrong.

I.e., for bounded inputs, the output is bounded.

$\therefore$ S1 is $\boxed{\text{stable}}$. (BIBO stable)

5. a) $\phi_{xy}(t) = \int_{-\infty}^{\infty} x(t+\tau)\, y(\tau)\, d\tau$ — ①

$\phi_{yx}(t) = \int_{-\infty}^{\infty} y(t+\tau)\, x(\tau)\, d\tau$ — ②

Let $\tau = r - t \Rightarrow d\tau = dr$; ② becomes:-

$\phi_{yx}(t) = \int_{-\infty}^{\infty} y(r)\, x(r-t)\, dr$

$= \int_{-\infty}^{\infty} x(-t+\tau)\, y(\tau)\, d\tau$ (dummy variable)

$= \phi_{xy}(-t)$

$\therefore \boxed{\phi_{yx}(t) = \phi_{xy}(-t)}$

b) $\phi_{xx}(t) = \int_{-\infty}^{\infty} x(t+\tau)\, x(\tau)\, d\tau$

From (a), $\phi_{yx}(t) = \phi_{xy}(-t)$

$\Rightarrow \phi_{xx}(t) = \phi_{xx}(-t)$

$\Rightarrow \phi_{xx}(t) \longrightarrow$ Even function

$\therefore$ Odd part of $\phi_{xx}(t) = \underline{\underline{0}}$.

c) $y(t) = x(t+T)$

$\phi_{xx}(t) = \int_{-\infty}^{\infty} x(t+\tau)\, x(\tau)\, d\tau$

- $\phi_{xy}(t) = \int_{-\infty}^{\infty} x(t+\tau)\, y(\tau)\, d\tau$

$= \int_{-\infty}^{\infty} x(t+\tau)\, x(\tau+T)\, d\tau$

Let $T+\tau = r \Rightarrow d\tau = dr$

$\Rightarrow \phi_{xy}(t) = \int_{-\infty}^{\infty} x(t-T+r)\, x(r)\, dr$

$= \int_{-\infty}^{\infty} x(\{t-T\}+\tau)\, x(\tau)\, d\tau$ [dummy variable]

$\therefore \boxed{\phi_{xy}(t) = \phi_{xx}(t-T)}$

- $\phi_{yy}(t) = \int_{-\infty}^{\infty} y(t+\tau)\, y(\tau)\, d\tau$

$= \int_{-\infty}^{\infty} x(t+\tau+T)\, x(\tau+T)\, d\tau$

Let $\tau+T = r \Rightarrow d\tau = dr$

$\Rightarrow \phi_{yy}(t) = \int_{-\infty}^{\infty} x(t+r)\, x(r)\, dr$

$= \int_{-\infty}^{\infty} x(t+\tau)\, x(\tau)\, d\tau$ [dummy variable]

$\therefore \boxed{\phi_{yy}(t) = \phi_{xx}(t)}$